

MLOps: Benefits, Applications and Popular Platforms



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MLOps is a fusion of DevOps and machine learning. It brings the principles that govern DevOps to the field of machine learning, to help build models that improve the functioning of industries as varied as healthcare and e-commerce. There are a number of platforms that can be used for MLOps. We take a look at a few of these.

Machine learning (ML) and DevOps are two critical fields that have been increasingly converging in recent years. DevOps refers to the specialised set of practices that combine software development (Dev) and IT operations (Ops) to shorten the systems development life cycle and provide continuous delivery with high software quality. When applied to ML, this convergence is often referred to as ‘MLOps’ (machine learning operations). MLOps aims to bring the principles of DevOps to the world of machine learning, facilitating

the seamless integration of ML models into production environments while ensuring reliability, scalability, and reproducibility.

Key advantages of implementing MLOps

Using MLOps offers several advantages that significantly improve the development, deployment and management of machine learning models. Through streamlined processes and enhanced reproducibility, MLOps empowers organisations to harness their full potential.

Faster development cycles: MLOps automates and streamlines the end-to-end machine learning life cycle, including model training, testing and deployment. This results in faster development cycles, allowing data scientists and engineers to iterate and improve models more rapidly.

Improved model quality: MLOps promotes continuous integration and continuous deployment (CI/CD) practices for machine learning models. As a result, models can be tested thoroughly and deployed with greater reliability, leading to higher model quality and performance.